8 **(a)** A charged oil drop is positioned between two horizontal metal plates, as shown in Fig. 8.1.

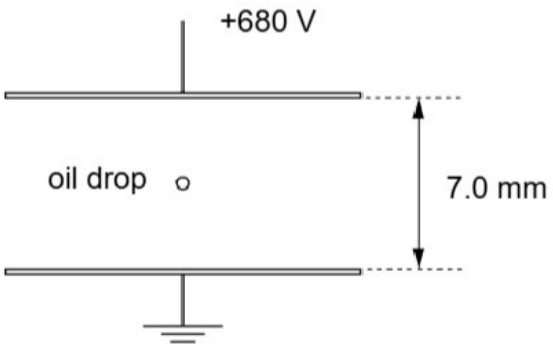


Fig. 8.1

The potential of the upper plate is gradually increased from zero till +680V, at which point the drop is held stationary.

The weight of the oil drop is found to be 4.8×10^{-14} N.

You may assume that upthrust to be negligible

(i) Deduce and explain the sign of the charge on the oil drop.

.....

.....

..... [2]

(ii) Hence, determine the charge on the oil drop.

charge = ... C [3]

(iii) Determine the potential difference required for the oil drop to accelerate upwards with an acceleration of 0.50 m s^{-2}.

potential difference = ... V [3]

(b) A particle of charge $+q$ and mass m enters a region of uniform magnetic field of flux density B of 9.7×10^{-2} T at a speed of 1.6×10^{5} m s^{-1} as shown in Fig. 8.2.

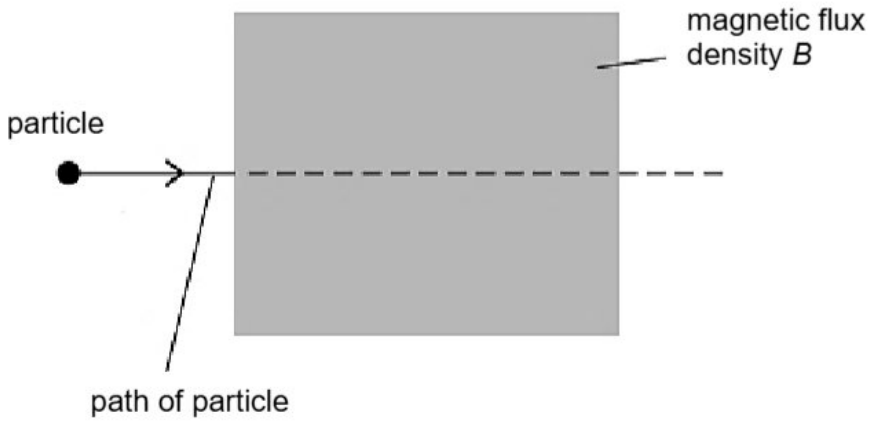


Fig. 8.2

A uniform electric field is applied in the same region so that the particle passes undeviated through the fields.

(i) The magnetic field is perpendicular to the plane of the page. On Fig. 8.2, draw arrows to indicate a possible combination of uniform magnetic and electric fields such that the particle can pass through undeviated. Label your fields with B and E respectively. [1]

(ii) Calculate the magnitude of the electric field strength.

electric field strength = ... $V\ m^{-1}$ [2]

(iii) A beam of particles with different speeds enters the crossed electric and magnetic fields.

By considering the forces on the particles, explain the effect of increasing the electric field strength on the speed of the particles that pass undeviated through the fields.

.....

.....

.....

..... [2]

(c) A horseshoe magnet is placed on a top pan balance. A rigid copper wire is fixed between the poles of the magnet, as illustrated in Fig. 8.3.

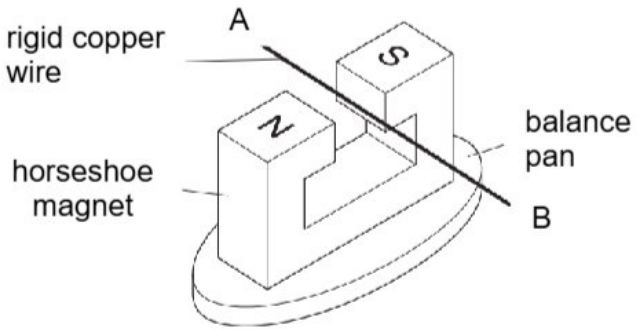


Fig. 8.3

(i) When a current is switched on, the balance reading decreases.

Explain why the balance reading decreases.

.....

.....

.....

..... [2]

(ii) When a direct current of 2.0 A flows in the wire, the balance reading becomes 2.20 N.

When the current is increased to 3.5 A, the balance reading decreases to 2.14 N.

1. Determine the mass of the magnet.

mass = ... kg [3]

2. State and explain the direction of the current in the wire.

.....

.....

.....

..... [2]